

COMMUNICATION AND INTERNET

Lesson-10

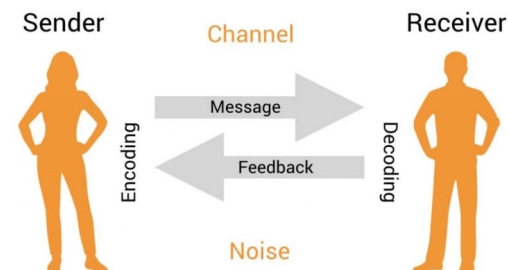
Class- Six

COMMUNICATION

- Communication is the act of transferring information from one place to another.
- It is a way of sending information from one computer or device to another.
- Communication can take place over a short or long distance.
- Modern communication can use:

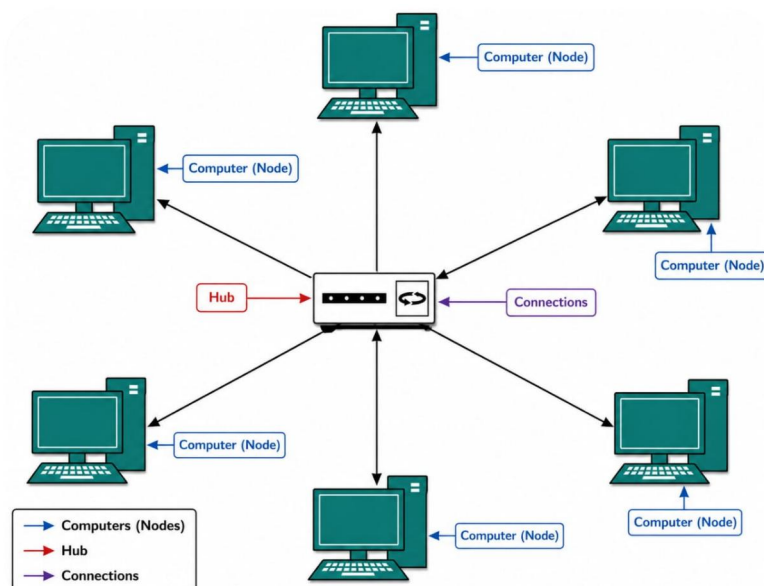
- Phones
- Satellites
- Radio links
- Computer
- The Internet

- Information communicated may include text, images, audio, Video and other data.
- Computer-based communication has made it possible to communicate with people anywhere in the world quickly.



COMPUTER NETWORK

- A computer network is a group of computers and other computing devices connected together.
- The connected devices communicate with each other through communication channels.
- Networks allow computers to exchange information and share resources.
- A computer network plays an important role in communication in modern digital society.
- Different types of networks are used according to the area and purpose.
- Through networking, users can share:
 - Files
 - Data
 - Printers
 - Programs
 - Other resources
- Computer networks make communication and resource sharing easier and faster.



INTERNET

- Internet stands for International Network.
- It is a worldwide network of interconnected computer networks.
- It connects millions of computers and other devices around the world.
- Computers connected to the Internet can communicate with each other.
- Users can send and receive:
 - Information
 - Data
 - Computer programs
 - Files
 - Other digital resources
- The Internet allows people to communicate and access information from different parts of the world.
- It is considered the world's largest computer network.

DEVELOPMENT OF COMPUTER NETWORK

- The development of computer networking began with the need to connect computers and share information.
- One of the most important early computer networks was ARPANET.
- ARPANET stands for Advanced Research Projects Agency Network.
- It was developed under the U.S. Department of Defense.
- ARPANET was established in the 1960s.
- It connected research organizations and computers so that they could communicate and share information.
- ARPANET became an important foundation for the development of the modern Internet.

Note: Our book specifically gives the history under “Who developed the Internet?”. So for notes, it is better to remember ARPANET + U.S. Department of Defense + 1960s rather than saying that one person invented the whole computer network.



BASIC REQUIREMENTS TO CONNECT INTERNET

The basic requirements mentioned in the book are:

❖ Modern Computer

A modern computer or suitable device is required.

It may be a:

- Desktop computer
- Laptop
- Or another suitable computing device.

❖ Telephone or Digital Cable Line

- A communication line may be required to connect the device to an Internet service.
- Telephone lines or digital cable connections can be used.

❖ Modulator and Demodulator

- A modulator and demodulator is commonly called a modem.
- It helps transmit and receive digital information through suitable communication connections.

❖ Network Interface Card / Router

- A Network Interface Card (NIC) allows a computer to connect to a network.
- A router can connect devices to a network and help provide Internet access.

❖ Internet Service Provider

- ISP stands for Internet Service Provider.
- An ISP provides Internet access to users.
- Users need an Internet connection from an ISP to access online services.

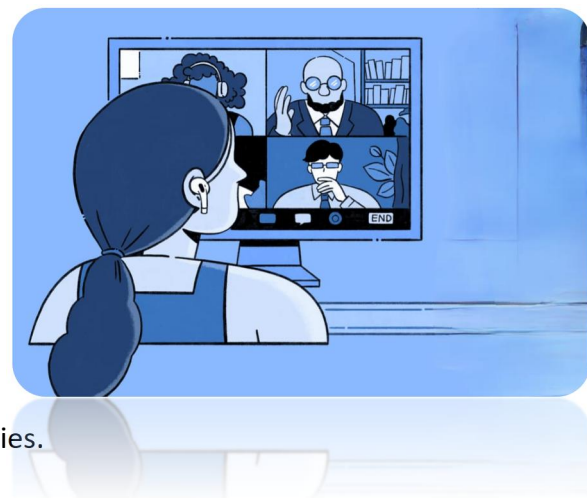
❖ Internet Browsing Software

- Browsing software is required to open and explore websites.
- Such software is called a web browser.
- Examples include Google Chrome and Mozilla Firefox.

USES OF INTERNET IN DIFFERENT PURPOSES

➤ Communication

- The Internet makes communication quick and easy.
- People can communicate with others who are far away.
- Users can communicate through:
 - Text messages
 - Online chats
 - Voice communication
 - Video calls
 - Video conferencing
- People can communicate with others in different countries.
- Internet communication saves both time and effort.
- It allows live communication between people in different locations.



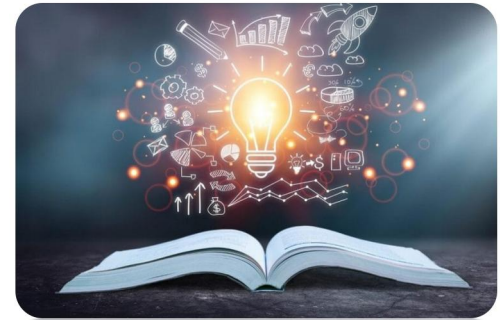
➤ Research

- The Internet is a very useful source of information for research.
- A large amount of information is available online.
- Information can be found with just a few clicks.
- Students and researchers can find references for their topics.
- It reduces the time required to search for information.
- Research can be done using:
 - Websites
 - Online books
 - Articles
 - Digital resources
 - Reference materials
- Therefore, the Internet has made the research process easier and faster.



➤ Education

- The Internet provides many resources for education and learning.
- Students can access online books and reference books.
- Educational websites provide information about different subjects.
- Students can find expert opinions and explanations online.
- Web-based training and online learning are possible.
- The Internet can make learning more interesting and easier.
- It provides resources that can support students' learning outside the classroom.
- Examples of educational resources include:
 - E-books
 - Online courses
 - Educational websites
 - Tutorials
 - Study materials



➤ Real-Time Updates

- The Internet provides information and updates in real time.
- Users can get current information from different parts of the world.
- It can provide updates about:
 - Business
 - Sports
 - Finance
 - Politics
 - Education
 - Entertainment
 - Information can be updated quickly as events happen.
 - Real-time updates help people stay informed about current situations.



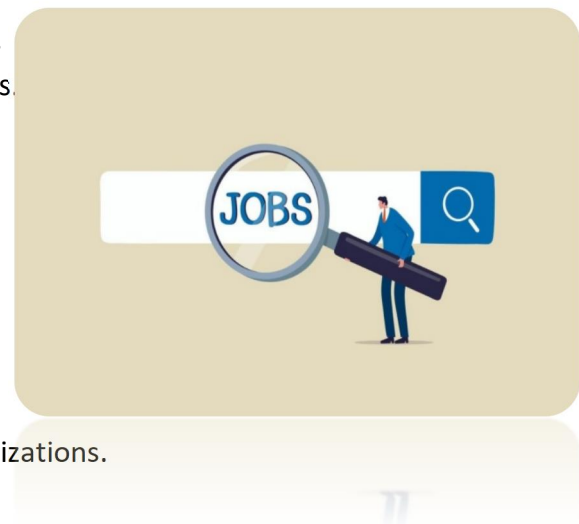
➤ Online Booking

- The Internet allows users to book different services online.
- Tickets for buses, trains and airplanes can be booked online.
- Online booking can be done from home.
- Users do not have to visit ticket counters.
- It saves:
 - Time
 - Travel
 - Effort
- Online booking is generally easier and more convenient.
- Users can select available services and make bookings online.



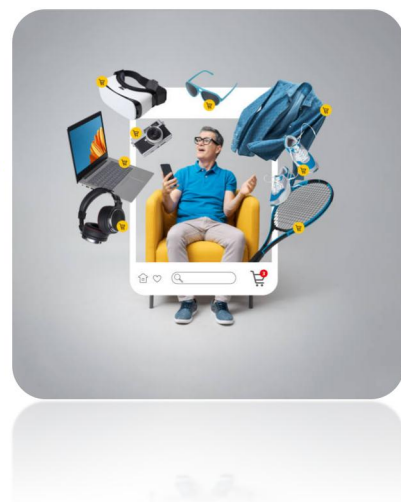
➤ Job Search

- The Internet has made searching for jobs easier.
- Many websites provide information about job vacancies.
- Users can search for jobs according to their requirements.
- Job websites may provide information about:
 - Vacancies
 - Organizations
 - Job requirements
 - Application procedures
- Users can register on job websites.
- They can apply for suitable vacancies online.
- Therefore, the Internet helps both job seekers and organizations.



➤ Online Shopping

- Online shopping allows people to buy products through the Internet.
- Users can visit online shopping websites from their computers or other devices.
- They can search for different products and compare available choices.
- Products can be ordered without visiting a physical shop.
- Online shopping can save time and effort.
- Products can be purchased from different places, including other parts of the world.
- Users can have products delivered to their desired location.



DIFFERENT TERMS USED ON THE INTERNET

➤ World Wide Web (WWW)

- WWW stands for World Wide Web.
- It is also called W3.
- The WWW is a collection of information and resources available through the Internet.

- It contains:
 - Text
 - Pictures
 - Sound
 - Videos
 - Multimedia
 - Hyperlinks
- These resources are linked and connected together.
- Users normally access the World Wide Web using a web browser.
- The WWW is one of the most widely used services of the Internet.



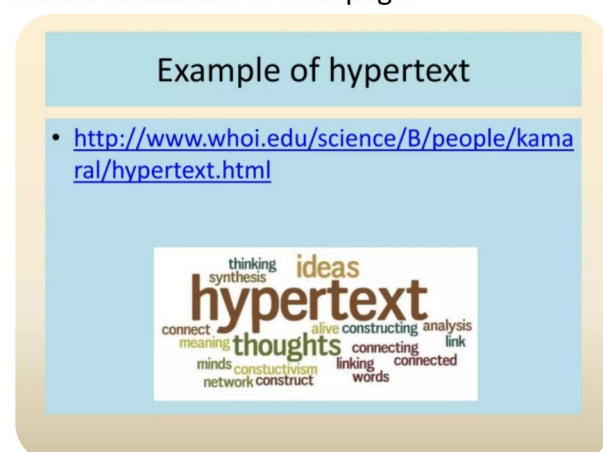
➤ Hyper Text Markup Language (HTML)

- HTML stands for Hyper Text Markup Language.
- HTML is a markup/coding language used to create web pages.
- It tells the web browser how different content should be displayed.
- HTML can be used to place:
 - Text
 - Pictures
 - Links
 - Other elements
- Web pages are commonly created using HTML.
- HTML provides the basic structure of a web page.



➤ Hypertext

- Hypertext is text on a web page that contains a link to another document or web page.
- A hypertext link may appear as:
 - A letter
 - A word
 - A phrase
 - A larger section of text
- When a user clicks the link, the browser can open the linked page or document.
- Hypertext makes it easy to move from one piece of information to another.



➤ Web Page

- A web page is a single page of information available on the Web.
- It can contain different types of content.
- A web page may contain:
 - Plain text

- Images
- Sound
- Videos
- Hyperlinks
- Other multimedia

- Web pages are commonly created using HTML.
- Several related web pages together can form a website.

➤ Website

- A website is a collection of related web pages.
- It has a unique Internet address.
- A website provides information or services to users.
- Websites can belong to:
 - Businesses
 - Schools
 - Colleges
 - Government offices
 - Organizations
 - Individuals
- A website normally contains a home page along with other web pages.
- Users can move between pages using links and menus.



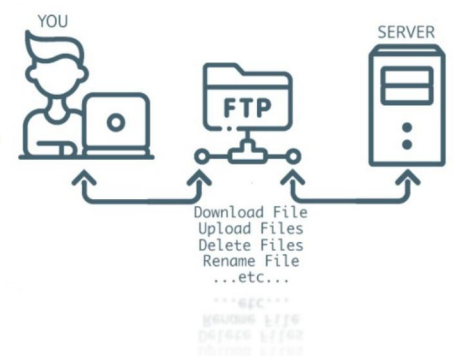
➤ Home Page

- A home page is the main or index page of a website.
- It is generally the starting point of a website.
- It provides links to other sections or pages of the website.
- When a user enters a website's main address, the home page is commonly displayed first.
- It helps users navigate through the website.
- In simple words, the home page is like the main entrance of a website.

FILE TRANSFERING AND BROWSING

➤ File Transfer Protocol (FTP)

- FTP stands for File Transfer Protocol.
- It is used to transfer files between computers over the Internet.
- It can transfer files between a user's computer and a server.
- FTP software helps make file transfer possible.
- Different types of files can be transferred, such as:
 - Documents
 - Pictures



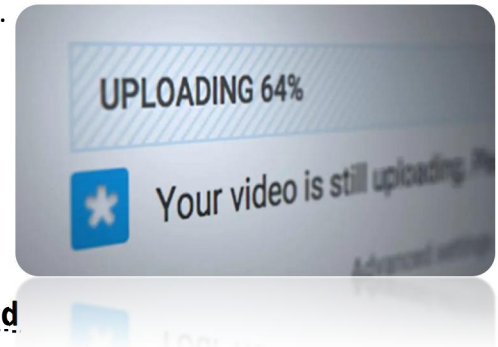
- Programs
 - Software
 - Other data
- FTP can be used for both uploading and downloading files.

➤ Uploading

- Uploading means transferring data or files from your computer to a server or another computer.
- The uploaded file is stored on the destination computer/server.
- Examples:
 - Uploading a photo to a website.
 - Uploading a document to an online system.
 - Uploading a file to a server.

Easy way to remember:

Your device → Internet/Server = Upload



➤ Downloading

- Downloading means transferring data or files from an Internet server or another computer to your computer/device.
- Downloaded files are stored on your device.
- Examples:
 - Downloading a picture.
 - Downloading a PDF.
 - Downloading software.
 - Downloading a document.

Easy way to remember:

Internet/Server → Your device = Download



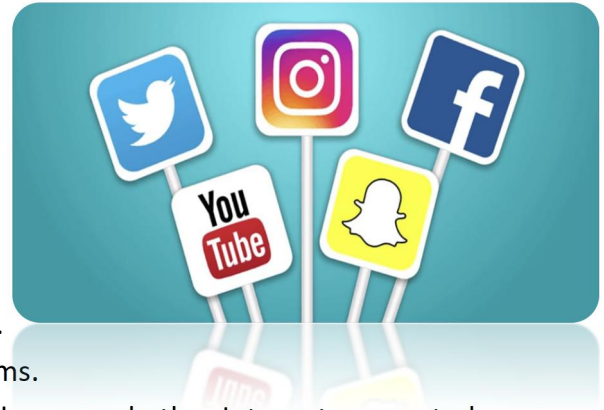
➤ Web Browser

- A web browser is a software application used to access and explore websites.
- It allows users to search for information on the Web.
- It can open and display web pages.
- Users enter a website address in the browser's address bar.
- Examples shown in the book include:
 - Mozilla Firefox
 - Google Chrome
 - Opera
 - Internet Explorer
 - Safari
- A browser is therefore an important tool for using the World Wide Web.

BASIC CONCEPT OF SOCIAL MEDIA

➤ Social Media

- Social media refers to computer-based technology that allows people to share information through virtual networks and communities.
- It allows users to communicate and share:
 - Ideas
 - Thoughts
 - Information
 - Documents
 - Videos and Photos
- Social media provides quick electronic communication.
- People can interact with others through online platforms.
- Social media can be accessed using computers, smartphones and other Internet-connected devices.



➤ Social Networking

- Social networking refers to using Internet-based social media sites to stay connected with other people.
- Social networking can have a social purpose, business purpose, or both.
- Users can communicate, share content and interact with other people.
- It can be used to stay connected with:
 - Friends
 - Family
 - Customers
 - Other users



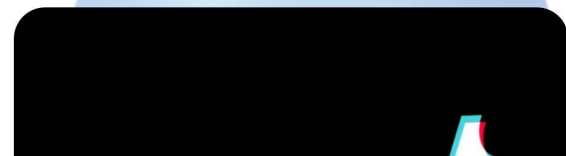
Examples are discussed in the book include Facebook and TikTok.

❖ Facebook

- Facebook is a social networking site.
- It helps people connect and share information with family and friends online.
- It was originally designed for college students.
- It was developed in 2004.
- Users can communicate and share different types of content.
- Facebook became one of the world's largest social networks.
- It can be used for communication and social interaction.



❖ TikTok



- TikTok is a social media application focused mainly on short videos.
- It is a social media and networking platform.
- Users can create and share short videos.
- Users can also watch videos created by other people.
- It has users from many countries around the world.
- TikTok is particularly popular for short-form video content.